AI-Based Load Forecasting for Microgrids

Godishala 33/11 kV Substation, Telangana, India

This report summarizes the leakage-safe Random Forest short-term load forecasting system developed from one complete year of hourly Indian substation demand and weather data.

Dataset Summary

Item	Value
Rows	8760
Start	2021-01-01T00:00:00
End	2021-12-31T23:00:00
Hours	8760
Missing Values	0
Load Mean Kw	2129.986
Load Min Kw	412.034
Load Max Kw	6306.206
Temperature Mean C	27.325
Humidity Mean Pct	67.726
Shutdown Hours	66
Iqr Outliers Replaced	0

Model Performance

Model	MAE (kW)	RMSE (kW)	MAPE (%)	R2
Random Forest	128.96	290.59	7.56	0.9171
Linear Regression	188.10	325.34	11.66	0.8961
24-hour Seasonal Naive	206.25	370.03	12.07	0.8656
ARIMA (2,1,0)	805.52	1023.18	53.11	-0.0277

Future Forecast

Forecast period: 2022-01-01 00:00:00 to 2022-01-01 23:00:00. Peak forecast: 4686.85 kW; average forecast: 2854.17 kW.

Timestamp	Forecast load (kW)	Temperature (C)	Humidity (%)
2022-01-01 00:00	1898.44	20.0	94.0
2022-01-01 01:00	1945.77	20.0	94.0
2022-01-01 02:00	1988.81	20.0	94.0
2022-01-01 03:00	2065.27	23.9	66.0

Timestamp	Forecast load (kW)	Temperature (C)	Humidity (%)
2022-01-01 04:00	2584.68	23.9	66.0
2022-01-01 05:00	3246.82	23.9	66.0
2022-01-01 06:00	3987.98	27.2	50.0
2022-01-01 07:00	4473.15	27.2	50.0
2022-01-01 08:00	4686.85	27.2	50.0
2022-01-01 09:00	4666.88	25.6	52.0
2022-01-01 10:00	4616.13	25.6	52.0
2022-01-01 11:00	4638.36	25.6	52.0
2022-01-01 12:00	4499.07	22.2	80.0
2022-01-01 13:00	4329.61	22.2	80.0
2022-01-01 14:00	4203.72	22.2	80.0
2022-01-01 15:00	4164.15	20.6	91.0
2022-01-01 16:00	2179.06	20.6	91.0
2022-01-01 17:00	1796.69	20.6	91.0
2022-01-01 18:00	1236.73	19.4	92.0
2022-01-01 19:00	1132.36	19.4	92.0
2022-01-01 20:00	1085.01	19.4	92.0
2022-01-01 21:00	1030.55	18.3	94.0
2022-01-01 22:00	995.67	18.3	94.0
2022-01-01 23:00	1048.34	18.3	94.0

Methodology Note

Missing values are forward-filled, IQR outliers are removed from the signal and replaced without deleting hourly timestamps, and MinMaxScaler is fitted only on training data. Random Forest tuning uses TimeSeriesSplit before evaluation on a chronological holdout set. Lag and rolling statistics use only past load values. Voltage, current, and power factor were excluded because they directly determine active power and would leak the prediction target.